

Healthcare Assistant
AI-Powered
• Generate Reports
• predict Diseases
• Answer patient Queries

Data Sources

- EHG/patient Records
- Lab Results
- Medical Images (X-Ray, CT)
- Clinical notes and prescriptions
- Wearable and vitals
- Research papers and Guidelines
- patient history

Artificial Intelligence (AI)

- The umbrella field focused on building intelligent systems that can perceive, reason, learn and make decisions.
- Overall goal : Build intelligent systems for healthcare.

Machine Learning (ML)

- A subset of AI that enables systems to learn patterns from data and make predictions or decisions without explicit programming
- Improves performance as more data becomes available.

Deep Learning

- Subset of ML using neural networks to learn complex patterns from large amounts of data.
- Handles complex data like medical images, ECGs, signals, and unstructured text.

Generative AI (Gen AI)

- Subset of AI focused on generating new content such as text, images, reports, summaries, or synthetic data.
- Can create synthetic data to augment training data.



Data Preprocessing and integration



CROSS-CUTTING ENABLERS
Privacy, security and compliance



Model Training, validation and Tuning



Deployment and Monitoring



Earlier and Accurate Diagnosis



Personalized Treatment

Outcomes



Improved efficiency and productivity



Better patient experience

Healthcare Assistant Applications

Predict Diseases

- predict risk of diseases and support early diagnosis and interventions.



Generate Medical Reports

- create clinical summaries, radiology reports, discharge summaries and recommendations



Answer patient queries

- provide accurate answers in language with 90%

